Mitchell B Slapik

MD/PhD Candidate & NIH Fellow, McGovern Medical School, mslapik@gmail.com

Education

2027	McGovern Medical School, *Houston, TX* Medical Scientist Training Program	MD
2025	Graduate School of Biomedical Sciences *Houston, TX,* Neuroscience	PhD
2017	Johns Hopkins University, *Baltimore, MD* Post-Baccalaureate Premedical Program	
2014	Swarthmore College, *Swarthmore, PA* With High Honors in Philosophy and Linguistics	BA
2013	University of Oxford, *Oxford, UK* Study Abroad: Philosophy of Mind	

Publications

2026

X. Niu, **M. Slapik**, A. McConnell et al. "Prefrontal Cortex Modulates Motor Encoding during Natural Behavior." In preparation.

J. Kim, M. Franch, M. Slapik et al. "Social Learning Refines dlPFC Context Encoding by Selectively Recruiting Context-informative Neurons." In preparation.

R. Milton*, **M. Slapik***, S. Egranov, et al. "Locomotor Activity Enhances Environmental Representation by Boosting Visuo-frontal Communication." Under review at *Nature Neuroscience*.

M. Slapik, H. Shouval. "Simulated Complex Cells Contribute to Object Recognition through Representational Untangling." *Neural Computation*.

2025

M. Slapik. "Computational Evidence for an Inverse Relationship between Brain and Retinal Complexity." *Journal of Vision*.

2022

M. Joyce, P. Nadkarni, S. Kronemer, [...], **M. Slapik**, et al. "Quality of Life Changes Following the Onset of Cerebellar Ataxia: Symptoms and Concerns Self-reported by Ataxia Patients and Informants." *Cerebellum*.

2020

O. Morgan, **M. Slapik**, K. Iannuzzelli, et al. "The Cerebellum and Sequencing in Motor and Cognitive Domains: Evidence from Cerebellar Ataxia." *Cerebellum*.

S. Kronemer, **M. Slapik**, J. Pietrowski, et al. "Neuropsychiatric Symptoms as a Reliable Phenomenology of Cerebellar Ataxia." *Cerebellum*.

2018 **M. Slapik**, S. I. Kronemer, O. Morgan, et al. "Visuospatial Organization and Recall in Cerebellar Ataxia." *Cerebellum*.

Research experience

2021 – Now **Graduate Research Assistant, Dragoi Lab**
McGovern Medical School, *Houston, TX*

Investigate parallels between biological and artificial intelligence in visual processing

- Analyzed neural activity in V4 and dlPFC during free exploration, and quantified inter-areal communication using reduced-rank regressions and spike-timing coordination
- Studied how early visual computations contribute to representational untangling in biological and artificial networks.
- Investigated the relationship between retinal and brain complexity using convolutional neural networks
- Designed optimal stimuli for neurons in visual cortex using an image generator and optimizer

2016 - 2019 **Research Assistant, Marvel Lab**
Johns Hopkins School of Medicine, *Baltimore, MD*

Examined the cognitive and emotional effects of cerebellar ataxia

- Designed new cognitive tasks assessing visuospatial skills, gestalt processing, sequence learning and verbal encoding
- Administered cognitive tasks, emotional questionnaires and motor tests to ataxia patients and controls
- Analyzed behavioral and questionnaire data, contributed to peer-reviewed manuscripts and presented findings to researchers, clinicians and patient communities

Fellowships and awards

2023 **National Research Service Award (F30)**
National Institutes of Health: National Eye Institute

2022 **Clinical and Translational Science Predoctoral Fellowship (TL1)**
National Institutes of Health: National Center for Advancing Translational Sciences

2024 **Dean's Research Scholarship**
McGovern Medical School

2023	**George M. Stancel Fellowship in the Biomedical Sciences** McGovern Medical School
2023	**Best Work-In-Progress Talk** Department of Neurobiology and Anatomy, McGovern Medical School
2022, 2023, 2024	**Osborne Endowed Scholarship in the Neurosciences** Department of Neurobiology and Anatomy, McGovern Medical School
2018	**Travel Award** for 8th Annual Ataxia Investigator's Meeting The National Ataxia Foundation
2014	**High Honors** in Philosophy and Linguistics Swarthmore College Honors Program

Presentation

Selected talks

2024	**M. Slapik**. "Retinal Complexity Varies Inversely with Brain Complexity in a Computational Model." Post-candidacy Talk at Fall Neuroscience Retreat for UTHealth. Houston, TX.
2023	**M. Slapik**, H. Shouval. "Unshattering Dimensionality." Post-candidacy Talk at Fall Neuroscience Retreat for UTHealth. Cleveland, TX.
2022	**M. Slapik**, A. Andrei, S. Khan, et al. "Optimal Stimuli as a New Method to Investigate Neural Networks." Pre-candidacy Talk at Fall Neuroscience Retreat for UTHealth. Cleveland, TX.
	M. Slapik, O. Morgan, J. Creighton, et al. "Timing and Sequencing in Cerebellar Ataxia." Nanosymposium talk accepted for presentation at: Society for Neuroscience San Diego, CA.
	M. Slapik, O. Morgan, C. Marvel. "Language Abilities in Cerebellar Ataxia." Presentation to the faculty and staff of the Johns Hopkins Ataxia Clinic, Baltimore, MD.
2017	**M. Slapik**, S. Kronemer, O. Morgan, et al. "Visuospatial Organization and Recall in Cerebellar Ataxia." Talk presented at: Sensorimotor Day, Johns Hopkins University, Baltimore, MD.

Selected posters

2024	**M. Slapik**, X. Niu, A. McConnell, et al. "Executive Regions Encode Movement of Individual Body Parts in Macaque Monkeys." Society for Neuroscience. Chicago, IL.

M. Slapik, H. Shouval. "The Visual System Relies on Organized but Low-Dimensional Representations for Object Recognition." Spring Neuroscience Retreat for UTHealth, Houston, TX.

2023 **M. Slapik**, A. Andrei, S. Khan, et al. "A Deep Learning Approach to Naturalistic Surround Modulation." Society for Neuroscience. Washington, D.C.

2022 **M. Slapik**, A. Andrei, S. Khan, et al. "Deep Networks Design Optimal Stimuli for Early Visual Cortex." Society for Neuroscience. San Diego, CA.

2020 **M. Slapik**, S. Patwardhan, R. Costa, et al. "Using Machine Learning To Classify Feeding Behavior in Aplysia." American Physician Scientists Association. Houston, TX.

2018 **M. Slapik**, J. Pietrowski, O. P. Morgan, et al. "A Characterization of Language Impairment in Cerebellar Ataxia." The National Ataxia Foundation's 8th Ataxia Investigator's Meeting. Philadelphia, PA.

2017 **M. Slapik**, S. Kronemer, J. Mandel, et al. "Visuospatial Processing and Strategy Formation in Cerebellar Ataxia." Society for Neuroscience, Washington, D.C.

Clinical and community service

2026 - Now **Curriculum Developer,** AI Standardized Patient Project
McGovern Medical School, *Houston, TX*

- Developed an AI-based standardized patient simulation for third-year medical students to practice psychiatric interviewing, diagnostic assessment, and clinical reasoning.
- Collaborated with psychiatry faculty to design case scenarios, learning objectives, and structured feedback for integration into the MS3 psychiatry curriculum.

2021 – Now **Volunteer Counselor**
Crisis Text Line, *Houston, TX*

- Provide crisis support via text to individuals experiencing suicidal ideation, self-harm urges, anxiety, depression, and acute psychosocial stressors
- Connect texters to personalized resources related to depression, anxiety, substance use, housing insecurity, and gender/sexual identity

2017 – 2019 **Team Leader, Health Resource Coordinator**
Charm City Clinic, *Baltimore, MD*

- Assisted clients with a variety of challenges related to medical care, employment, and housing
- Trained a new group of volunteers

2015 – 2016 Emergency Room Volunteer
Penn Presbyterian Medical Center, *Philadelphia, PA*
- Took incoming calls, paged nurses, restocked supplies, and observed procedures

2015 – 2016 Front-Desk Volunteer
Washington West Project, *Philadelphia, PA*
- Enrolled patients for STD screening and counseling

Research service

Mentoring
Xiaoke Niu, Jaehwan Kim, Joey Zambelas, Madeline Gomez

Reviewing
Scientific Reports, Frontiers in Computational Neuroscience, Cerebellum

Machine Learning Journal Club
Leadership Committee

Certificates

2023 Machine Learning Specialization
Stanford University (Coursera)

Deep Learning Specialization
deeplearning.ai (Coursera)

Organizations

2017 - Now Society for Neuroscience

2017 - 2019 National Ataxia Foundation

Skills

Electrophysiology: Acute and chronic recording, spike-sorting, eye tracking, receptive field mapping, behavioral task design

MRI: Structural scans

Data analysis: MATLAB, Python, SPSS

Machine Learning: PyTorch, scikit-learn, dimensionality reduction, decoding, regression

Task Development: E-Prime, PsychToolbox, MonkeyLogic

Languages

English: Fluent

Spanish: Basic

Interests

Jazz Saxophone
The Chirp Chirps, Bayou City Funk

Classical and Jazz Piano